20261015 Quiz

for (MATH102) Calculus II

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Problem 3.15. Find an equation for the plane tangent to the graph of $f(x, y) = x + y^2$,

(a) at $(x, y) = (0, 0)$,
(b) at $(x, y) = (1, 2)$.

Solution. Since $f_x(x, y) = 1$ and $f_y(x, y) = 2y$, the tangent plane at $(a, b, f(a, b))$ is

$$z = f(a, b) + f_x(a, b)(x - a) + f_y(a, b)(y - b).$$

(a) At $(0, 0)$, we have $f(0, 0) = 0$, $f_x(0, 0) = 1$, and $f_y(0, 0) = 0$. Hence

$$\boxed{z = x}.$$

(b) At $(1, 2)$, we have $f(1, 2) = 5$, $f_x(1, 2) = 1$, and $f_y(1, 2) = 4$. Thus

$$\boxed{z = 5 + (x - 1) + 4(y - 2)}.$$

□

Problem 3.16. Let

$$f(x, y) = \sqrt{1 - x^2 - y^2}.$$

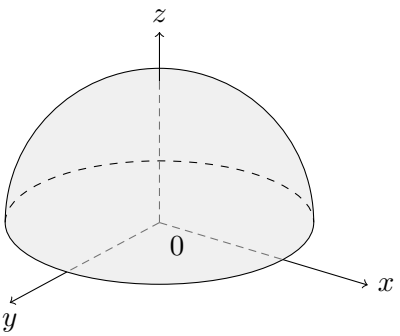
(a) Sketch the graph of f.
(b) Find an equation of the plane tangent to the graph of f at $(x, y) = (\sqrt{0.4}, \sqrt{0.5})$.

Solution.

(a) Since $z = f(x, y) \geq 0$ and

$$x^2 + y^2 + z^2 = 1,$$

the graph is the upper hemisphere of the unit sphere, including its boundary circle $x^2 + y^2 = 1$ in the plane $z = 0$.



(b) Put $a = \sqrt{0.4} = \sqrt{\frac{2}{5}}$ and $b = \sqrt{0.5} = \frac{1}{\sqrt{2}}$. Then

$$f(a,b) = \sqrt{1 - \frac{2}{5} - \frac{1}{2}} = \frac{1}{\sqrt{10}}.$$

In the interior of the unit disk,

$$f_x(x,y) = -\frac{x}{\sqrt{1-x^2-y^2}}, \qquad f_y(x,y) = -\frac{y}{\sqrt{1-x^2-y^2}}.$$

Consequently,

$$f_x(a,b) = -2, \qquad f_y(a,b) = -\sqrt{5}.$$

The tangent plane is therefore

$$z = \frac{1}{\sqrt{10}} - 2(x-a) - \sqrt{5}(y-b),$$

or, equivalently,

$$\boxed{2x + \sqrt{5}\,y + z = \sqrt{10}}.$$

□

Problem 3.18. Find the following derivatives or partial derivatives.

(a) $\dfrac{\mathrm{d}}{\mathrm{d}x}(f(x)^3)$, where $f\colon \mathbb{R} \to \mathbb{R}$ is differentiable.

(b) $\dfrac{\partial}{\partial x}(y + x^2)^3$.

(c) $\dfrac{\partial}{\partial x} g(y + x^2)$, where $g\colon \mathbb{R} \to \mathbb{R}$ is differentiable.

(d) $\dfrac{\partial}{\partial y}(f(x,y)^3)$, where $f\colon \mathbb{R}^2 \to \mathbb{R}$ is differentiable.

Solution. Applying the Chain Rule, with the other variable held fixed for partial derivatives, gives

(a)

$$\boxed{\frac{\mathrm{d}}{\mathrm{d}x}(f(x)^3) = 3f(x)^2 f'(x)}.$$

(b)

$$\boxed{\frac{\partial}{\partial x}(y + x^2)^3 = 6x(y + x^2)^2}.$$

(c)

$$\boxed{\frac{\partial}{\partial x} g(y + x^2) = 2x g'(y + x^2)}.$$

(d)

$$\boxed{\frac{\partial}{\partial y}(f(x,y)^3) = 3f(x,y)^2 f_y(x,y)}.$$

□

Problem 3.19. Find the derivatives f_{xx}, f_{xy}, and f_{yy} for each function.

(a) $f(x,y) = x^2 - y^2$.
(b) $f(x,y) = x^2 + y^2$.
(c) $f(x,y) = (x+y)^2$.
(d) $f(x,y) = e^{-x}\cos y$.
(e) $f(x,y) = e^{-ay}\sin(ax)$, where a is a constant.

Solution.

(a) Here $f_x = 2x$ and $f_y = -2y$, so

$$\boxed{f_{xx} = 2, \qquad f_{xy} = 0, \qquad f_{yy} = -2}.$$

(b) Here $f_x = 2x$ and $f_y = 2y$, so

$$\boxed{f_{xx} = 2, \qquad f_{xy} = 0, \qquad f_{yy} = 2}.$$

(c) Here $f_x = 2(x+y) = f_y$, so

$$\boxed{f_{xx} = 2, \qquad f_{xy} = 2, \qquad f_{yy} = 2}.$$

(d) Since

$$f_x = -e^{-x}\cos y, \qquad f_y = -e^{-x}\sin y,$$

we obtain

$$\boxed{f_{xx} = e^{-x}\cos y, \quad f_{xy} = e^{-x}\sin y, \quad f_{yy} = -e^{-x}\cos y}.$$

(e) Since

$$f_x = ae^{-ay}\cos(ax), \qquad f_y = -ae^{-ay}\sin(ax),$$

we obtain

$$\boxed{\begin{aligned} f_{xx} &= -a^2e^{-ay}\sin(ax), \\ f_{xy} &= -a^2e^{-ay}\cos(ax), \\ f_{yy} &= a^2e^{-ay}\sin(ax). \end{aligned}}$$

□

Problem 3.23. Let $f(x,y) = 4 + 3x + 2y + xy$ and $\mathbf{a} = (1,1)$. Find the directional derivatives

$$D_{\mathbf{v}}f(\mathbf{a}) = \mathbf{v} \cdot \nabla f(\mathbf{a})$$

in each of the following directions, in order:

$$(1,0), \quad \left(\frac{4}{5}, \frac{3}{5}\right), \quad (0,1), \quad \left(-\frac{3}{5}, \frac{4}{5}\right), \quad (-1,0).$$

Why is one of these derivatives equal to $\|\nabla f(\mathbf{a})\|$?

Solution. Since

$$\nabla f(x,y) = (3+y, 2+x), \qquad \nabla f(\mathbf{a}) = (4,3),$$

we evaluate $D_{\mathbf{v}}f(\mathbf{a}) = \mathbf{v} \cdot (4,3)$ in the given directions, obtaining

$$\boxed{4, \qquad 5, \qquad 3, \qquad 0, \qquad -4},$$

respectively. The second direction is the unit vector

$$\mathbf{v} = \left(\frac{4}{5}, \frac{3}{5}\right) = \frac{\nabla f(\mathbf{a})}{\|\nabla f(\mathbf{a})\|}.$$

Thus its directional derivative equals $\|\nabla f(\mathbf{a})\| = 5$.

□

Problem 3.29. Suppose $u\colon \mathbb{R}^2 \to \mathbb{R}$ is a C^2 function. After the change to polar coordinates

$$x = r\cos\theta, \qquad y = r\sin\theta,$$

write $U(r,\theta) = u(r\cos\theta, r\sin\theta)$. By the Chain Rule,

$$U_r = \frac{\partial u}{\partial x}\frac{\partial x}{\partial r} + \frac{\partial u}{\partial y}\frac{\partial y}{\partial r}, \qquad U_\theta = \frac{\partial u}{\partial x}\frac{\partial x}{\partial \theta} + \frac{\partial u}{\partial y}\frac{\partial y}{\partial \theta},$$

where the derivatives of u on the right are evaluated at (x,y). Use the Chain Rule to show that, for $r > 0$,

$$u_{xx} + u_{yy} = U_{rr} + \frac{1}{r}U_r + \frac{1}{r^2}U_{\theta\theta}.$$

Solution. By the Chain Rule,

$$U_r = u_x\cos\theta + u_y\sin\theta, \qquad U_\theta = -ru_x\sin\theta + ru_y\cos\theta.$$

Differentiating with respect to r gives

$$U_{rr} = u_{xx}\cos^2\theta + 2u_{xy}\sin\theta\cos\theta + u_{yy}\sin^2\theta.$$

Differentiating with respect to θ gives

$$\begin{aligned} U_{\theta\theta} &= -r(u_x\cos\theta + u_y\sin\theta) \\ &\quad + r^2(u_{xx}\sin^2\theta - 2u_{xy}\sin\theta\cos\theta + u_{yy}\cos^2\theta) \\ &= -rU_r + r^2(u_{xx}\sin^2\theta - 2u_{xy}\sin\theta\cos\theta + u_{yy}\cos^2\theta). \end{aligned}$$

For $r > 0$, adding U_{rr} and $r^{-2}(U_{\theta\theta} + rU_r)$ yields

$$\boxed{u_{xx} + u_{yy} = U_{rr} + \frac{1}{r}U_r + \frac{1}{r^2}U_{\theta\theta}},$$

since $\sin^2\theta + \cos^2\theta = 1$.

□

Problem 3.31. Let $u(x,y) = x^2 - y^2$ and $v(x,y) = 2xy$.

(a) Show that $u_x = v_y$ and $u_y = -v_x$.

(b) Show that $u_{xx} + u_{yy} = 0$.

(c) Define $w(x,y) = u(u(x,y), v(x,y))$. Show that $w_{xx} + w_{yy} = 0$.

(d) Suppose p, q, and r are C^2 functions of two variables satisfying

$$p_x = q_y, \qquad p_y = -q_x, \qquad r_{xx} + r_{yy} = 0.$$

Define $w(x,y) = r(p(x,y), q(x,y))$. Show that $w_{xx} + w_{yy} = 0$.

Solution.

(a) Direct differentiation gives

$$u_x = 2x = v_y, \qquad u_y = -2y = -v_x.$$

(b) Since $u_{xx} = 2$ and $u_{yy} = -2$,

$$\boxed{u_{xx} + u_{yy} = 0}.$$

(c) We have

$$w = u^2 - v^2 = (x^2 - y^2)^2 - (2xy)^2 = x^4 - 6x^2y^2 + y^4.$$

Hence

$$w_{xx} = 12x^2 - 12y^2, \qquad w_{yy} = -12x^2 + 12y^2,$$

so

$$\boxed{w_{xx} + w_{yy} = 0}.$$

(d) Write $r = r(s,t)$ and evaluate its derivatives at $(s,t) = (p(x,y), q(x,y))$. Twice applying the Chain Rule yields

$$\begin{aligned} w_{xx} + w_{yy} = {} & (p_x^2 + p_y^2) r_{ss} + 2(p_x q_x + p_y q_y) r_{st} \\ & + (q_x^2 + q_y^2) r_{tt} + r_s(p_{xx} + p_{yy}) + r_t(q_{xx} + q_{yy}). \end{aligned}$$

By $p_x = q_y$ and $p_y = -q_x$,

$$p_x q_x + p_y q_y = 0, \qquad p_x^2 + p_y^2 = q_x^2 + q_y^2.$$

Also, equality of mixed partial derivatives gives

$$p_{xx} + p_{yy} = q_{yx} - q_{xy} = 0, \qquad q_{xx} + q_{yy} = -p_{yx} + p_{xy} = 0.$$

Therefore

$$w_{xx} + w_{yy} = (p_x^2 + p_y^2)(r_{ss} + r_{tt}) = 0,$$

because $r_{ss} + r_{tt} = 0$. Hence $\boxed{w_{xx} + w_{yy} = 0}$.

□

Problem 3.35. Let

$$\mathbf{F}(x,y) = (e^x \cos y, e^x \sin y).$$

(a) Find the derivative matrix $D\mathbf{F}$. That is, writing $u = e^x \cos y$ and $v = e^x \sin y$, find

$$\begin{pmatrix} \dfrac{\partial u}{\partial x} & \dfrac{\partial u}{\partial y} \\ \dfrac{\partial v}{\partial x} & \dfrac{\partial v}{\partial y} \end{pmatrix}.$$

(b) Show that $\big(D\mathbf{F}(\mathbf{a})\big)^{-1}$ exists at every point $\mathbf{a} = (a,b)$, so $\mathbf{F}$ is locally invertible at $\mathbf{a}$.

(c) Show that $\mathbf{F}$ is not invertible globally; that is, find different points of the domain that are mapped to the same value in the range.

Solution.

(a) Differentiating both components gives

$$\boxed{D\mathbf{F}(x,y) = \begin{pmatrix} e^x \cos y & -e^x \sin y \\ e^x \sin y & e^x \cos y \end{pmatrix}}.$$

(b) The Jacobian determinant is

$$\boxed{\det D\mathbf{F}(a,b) = e^{2a} > 0}.$$

Hence $D\mathbf{F}(\mathbf{a})$ is invertible at every point. By the Inverse Function Theorem (Theorem 3.11), $\mathbf{F}$ is locally invertible at every $\mathbf{a} \in \mathbb{R}^2$.

(c) Since

$$\mathbf{F}(x, y + 2\pi) = \mathbf{F}(x, y)$$

for every (x, y), distinct domain points have the same image. Thus $\mathbf{F}$ is not injective and has no global inverse on $\mathbb{R}^2$.

□

Problem 3.36. Let $\mathbf{F}(x, y) = (x \cos y, x \sin y)$ for $x > 0$.

(a) Find the derivative matrix $D\mathbf{F}(\mathbf{a})$ at $\mathbf{a} = (a, b)$.

(b) Show that $\big(D\mathbf{F}(\mathbf{a})\big)^{-1}$ exists at every point with $a > 0$, so $\mathbf{F}$ is locally invertible at $\mathbf{a}$.

Solution.

(a) Differentiating both components gives

$$\boxed{D\mathbf{F}(a, b) = \begin{pmatrix} \cos b & -a \sin b \\ \sin b & a \cos b \end{pmatrix}}.$$

(b) We have

$$\boxed{\det D\mathbf{F}(a, b) = a(\cos^2 b + \sin^2 b) = a > 0}.$$

Thus $D\mathbf{F}(\mathbf{a})$ is invertible for $a > 0$. By the Inverse Function Theorem, $\mathbf{F}$ is locally invertible at every point in its domain.

□

Problem 3.37. Let $\mathbf{F}(x, y) = (x^4 + 2xy + 1, y)$, and consider the equations

$$x^4 + 2xy + 1 = u, \qquad y = v,$$

that is, $\mathbf{F}(x, y) = (u, v)$.

(a) Show that for $(u, v) = (0, 1)$, one solution is $(x, y) = (-1, 1)$.

(b) Find the derivative matrix $D\mathbf{F}(x, y)$.

(c) Show that $D\mathbf{F}(-1, 1)$ is invertible, and hence $\mathbf{F}^{-1}$ exists on a small enough disk centered at $(0, 1)$.

(d) Use the linear approximation to estimate $\mathbf{F}^{-1}(0.2, 1.01)$.

Solution.

(a) Substitution gives

$$(-1)^4 + 2(-1)(1) + 1 = 0, \qquad y = 1,$$

so $\boxed{\mathbf{F}(-1, 1) = (0, 1)}$.

(b) The derivative matrix is

$$\boxed{D\mathbf{F}(x, y) = \begin{pmatrix} 4x^3 + 2y & 2x \\ 0 & 1 \end{pmatrix}}.$$

(c) At $(-1, 1)$,

$$D\mathbf{F}(-1, 1) = \begin{pmatrix} -2 & -2 \\ 0 & 1 \end{pmatrix}, \qquad \det D\mathbf{F}(-1, 1) = -2 \neq 0.$$

By the Inverse Function Theorem, a local inverse exists near $(0, 1)$.

(d) The derivative of the local inverse at $(0,1)$ is

$$D\mathbf{F}^{-1}(0,1) = (D\mathbf{F}(-1,1))^{-1} = \begin{pmatrix} -\frac{1}{2} & -1 \\ 0 & 1 \end{pmatrix}.$$

The linear approximation gives the coordinate increments

$$\begin{pmatrix} \Delta x \\ \Delta y \end{pmatrix} \approx D\mathbf{F}^{-1}(0,1) \begin{pmatrix} 0.2 \\ 0.01 \end{pmatrix} = \begin{pmatrix} -0.11 \\ 0.01 \end{pmatrix}.$$

Adding these increments to $(-1,1)$ yields

$$\boxed{\mathbf{F}^{-1}(0.2, 1.01) \approx (-1.11, 1.01)}.$$

□

Problem 3.40. Let $f(x,y,z) = z^4 + 2yz + x$, and consider the level set

$$z^4 + 2yz + x = 0.$$

(a) Show that $(0,2,0)$ is on the level set.
(b) Find the partial derivatives of f.
(c) Show that there is a function $z = g(x,y)$ defined near $(0,2)$ so that $f(x,y,g(x,y)) = 0$.
(d) Find the partial derivatives $g_x(x,y)$ and $g_y(x,y)$, and use them to find the second derivatives $g_{xx}(0,2)$, $g_{xy}(0,2)$, and $g_{yy}(0,2)$.

Solution.

(a) We have $f(0,2,0) = 0^4 + 2(2)(0) + 0 = 0$, as required.
(b) Direct differentiation gives

$$\boxed{f_x = 1, \qquad f_y = 2z, \qquad f_z = 4z^3 + 2y}.$$

(c) At $(0,2,0)$,

$$f_z(0,2,0) = 4 \neq 0.$$

Since f is smooth, the Implicit Function Theorem (Theorem 3.13) gives a C^2 function g defined near $(0,2)$ satisfying $g(0,2) = 0$ and $f(x,y,g(x,y)) = 0$.

(d) Differentiating

$$g(x,y)^4 + 2yg(x,y) + x = 0$$

with respect to x and y gives

$$(4g^3 + 2y)g_x + 1 = 0, \qquad (4g^3 + 2y)g_y + 2g = 0.$$

Hence, near $(0,2)$,

$$\boxed{g_x = -\frac{1}{4g^3 + 2y}, \qquad g_y = -\frac{2g}{4g^3 + 2y}}.$$

In particular, $g(0,2) = 0$, $g_x(0,2) = -\frac{1}{4}$, and $g_y(0,2) = 0$.
Differentiating the identity for g_x once more with respect to x and y, respectively, gives

$$(12g^2 g_x)g_x + (4g^3 + 2y)g_{xx} = 0,$$
$$(12g^2 g_y + 2)g_x + (4g^3 + 2y)g_{xy} = 0.$$

Likewise, differentiating the identity for g_y with respect to y yields

$$(12g^2 g_y + 2)g_y + (4g^3 + 2y)g_{yy} + 2g_y = 0.$$

Evaluating at $(0, 2)$, where $g = 0$, gives

$$\boxed{g_{xx}(0,2) = 0, \qquad g_{xy}(0,2) = \frac{1}{8}, \qquad g_{yy}(0,2) = 0}.$$

□

Problem 3.41. Let

$$f_1(x, y_1, y_2) = 3y_1 + y_2^2 + 4x = 0, \qquad f_2(x, y_1, y_2) = 4y_1^3 + y_2 + x = 0.$$

(a) Verify that $(x, y_1, y_2) = (-4, 0, 4)$ is a solution.

(b) Show that there is a function $\mathbf{G} = (g_1, g_2)$ such that $g_1(x) = y_1$ and $g_2(x) = y_2$ for all solutions near $(-4, 0, 4)$.

(c) Show that at $x = -4$,

$$\frac{\mathrm{d}g_1}{\mathrm{d}x} = \frac{4}{3}, \qquad \frac{\mathrm{d}g_2}{\mathrm{d}x} = -1.$$

Solution.

(a) Substituting $(-4, 0, 4)$ gives

$$f_1(-4, 0, 4) = 0 + 16 - 16 = 0, \qquad f_2(-4, 0, 4) = 0 + 4 - 4 = 0.$$

(b) The Jacobian with respect to (y_1, y_2) is

$$\begin{pmatrix} \dfrac{\partial f_1}{\partial y_1} & \dfrac{\partial f_1}{\partial y_2} \\ \dfrac{\partial f_2}{\partial y_1} & \dfrac{\partial f_2}{\partial y_2} \end{pmatrix} = \begin{pmatrix} 3 & 2y_2 \\ 12y_1^2 & 1 \end{pmatrix}.$$

At $(-4, 0, 4)$ this matrix is

$$\begin{pmatrix} 3 & 8 \\ 0 & 1 \end{pmatrix}, \qquad \det \begin{pmatrix} 3 & 8 \\ 0 & 1 \end{pmatrix} = 3 \neq 0.$$

Therefore the Implicit Function Theorem (Theorem 3.14) gives a local function $\mathbf{G} = (g_1, g_2)$ with $\mathbf{G}(-4) = (0, 4)$ solving both equations.

(c) Differentiating $f_1(x, g_1(x), g_2(x)) = 0$ and $f_2(x, g_1(x), g_2(x)) = 0$ gives

$$3g_1' + 2g_2 g_2' + 4 = 0, \qquad 12g_1^2 g_1' + g_2' + 1 = 0.$$

At $x = -4$, where $g_1 = 0$ and $g_2 = 4$, these become

$$3g_1'(-4) + 8g_2'(-4) = -4, \qquad g_2'(-4) = -1.$$

Thus

$$\boxed{g_1'(-4) = \frac{4}{3}, \qquad g_2'(-4) = -1}.$$

□

Problem 3.42. Let $\mathbf{F}(u,v,w) = (u+v+w, uv+vw+wu)$, and consider the system of equations $\mathbf{F}(u,v,w) = (2,-4)$:

$$u+v+w=2, \qquad uv+vw+wu=-4.$$

(a) Verify that $(2,2,-2)$ is a solution.

(b) Writing $\mathbf{F}(u,v,w) = (f_1(u,v,w), f_2(u,v,w))$, find all the first partial derivatives of f_1 and f_2.

(c) Verify that

$$D\mathbf{F}(2,2,-2) = \begin{pmatrix} 1 & 1 & 1 \\ 0 & 0 & 4 \end{pmatrix}.$$

(d) Can the equations be solved for v and w in terms of u near $(2,2,-2)$? That is, does the Implicit Function Theorem guarantee a function $\mathbf{G} = (g_1, g_2)$ on an interval around 2 such that $\mathbf{G}(2) = (2,-2)$ and

$$\mathbf{F}(u, g_1(u), g_2(u)) = (2,-4)?$$

(e) Can the equations be solved for u and v in terms of w near $(2,2,-2)$?

Solution.

(a) We calculate

$$2+2-2=2, \qquad (2)(2)+(2)(-2)+(-2)(2) = -4.$$

Thus $(2,2,-2)$ solves both equations.

(b) The component derivatives are

$$(f_{1,u}, f_{1,v}, f_{1,w}) = (1,1,1),$$
$$(f_{2,u}, f_{2,v}, f_{2,w}) = (v+w, u+w, u+v).$$

(c) Thus

$$D\mathbf{F}(u,v,w) = \begin{pmatrix} 1 & 1 & 1 \\ v+w & u+w & u+v \end{pmatrix}, \qquad \boxed{D\mathbf{F}(2,2,-2) = \begin{pmatrix} 1 & 1 & 1 \\ 0 & 0 & 4 \end{pmatrix}}.$$

(d) The (v,w)-Jacobian at $(2,2,-2)$ is

$$\begin{pmatrix} 1 & 1 \\ 0 & 4 \end{pmatrix}, \qquad \det \begin{pmatrix} 1 & 1 \\ 0 & 4 \end{pmatrix} = 4 \neq 0.$$

By the Implicit Function Theorem, a local function $\mathbf{G}(u) = (g_1(u), g_2(u))$ exists with $\mathbf{G}(2) = (2,-2)$ and $\mathbf{F}(u, g_1(u), g_2(u)) = (2,-4)$. Thus the answer is $\boxed{\text{Yes}}$.

(e) The (u,v)-Jacobian at $(2,2,-2)$ is

$$\begin{pmatrix} 1 & 1 \\ 0 & 0 \end{pmatrix},$$

which is singular, so the Implicit Function Theorem gives no guarantee. In fact, the equations imply $u+v = 2-w$ and $uv = w^2 - 2w - 4$. Real u, v require the discriminant

$$(u+v)^2 - 4uv = (w+2)(10-3w) \geq 0.$$

For $w < -2$ sufficiently close to -2, however, this quantity is negative. Since no real solutions exist at these nearby values, u and v cannot be defined as functions of w on an interval around -2. Thus the answer is $\boxed{\text{No}}$.

□